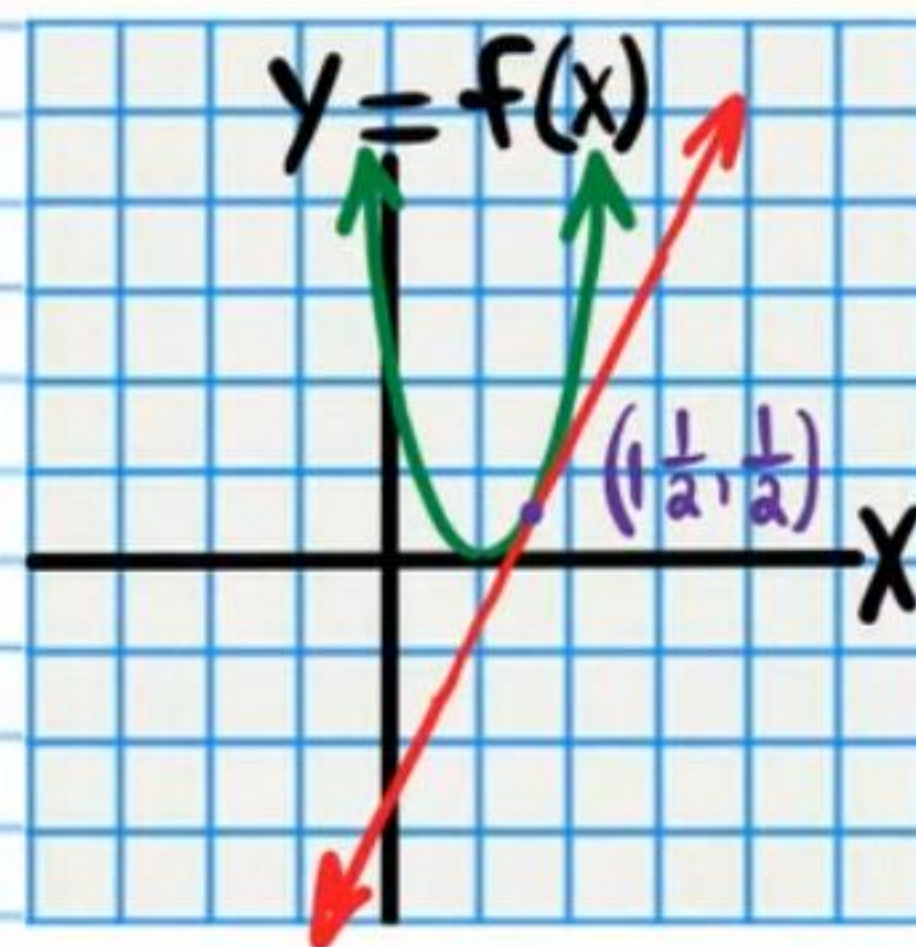


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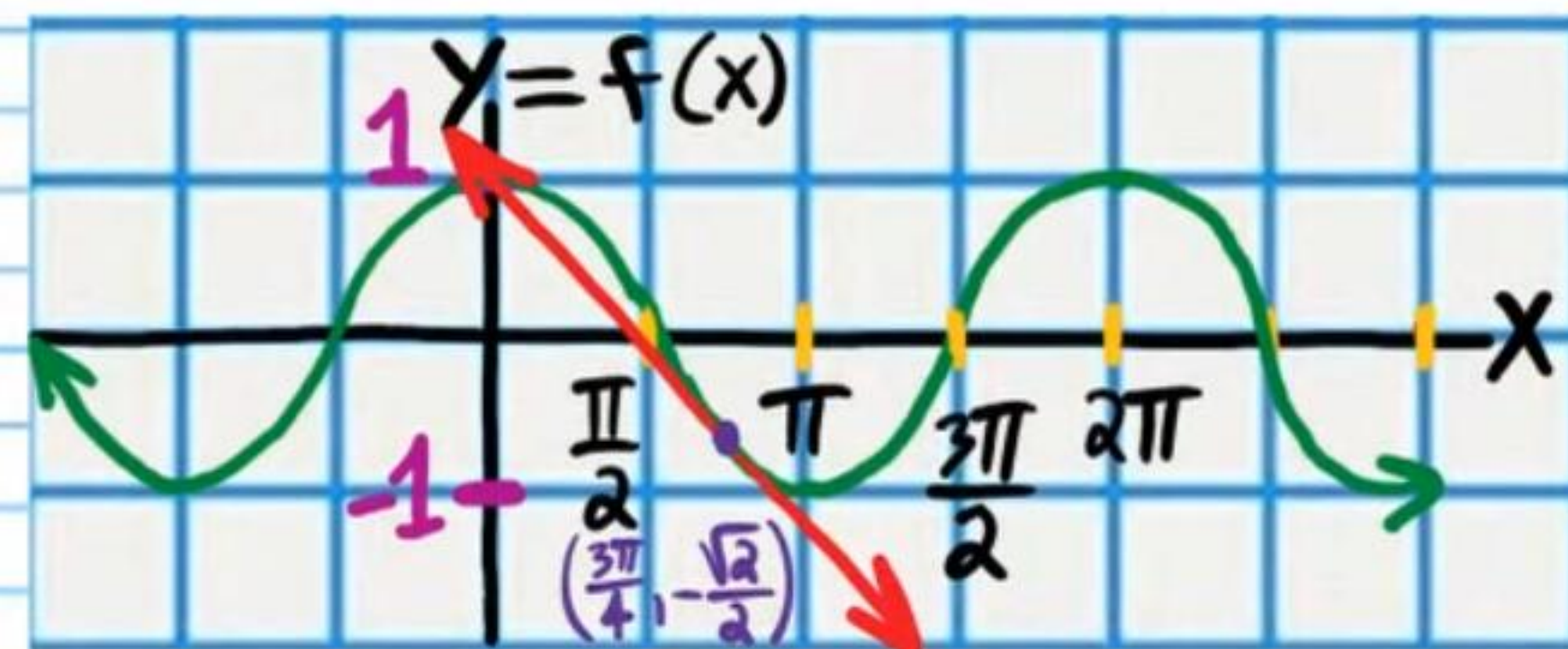
Finding the Equation of a Tangent Line (Homework)

Find an equation of the line tangent to the graph of each function at the indicated point on the graph.

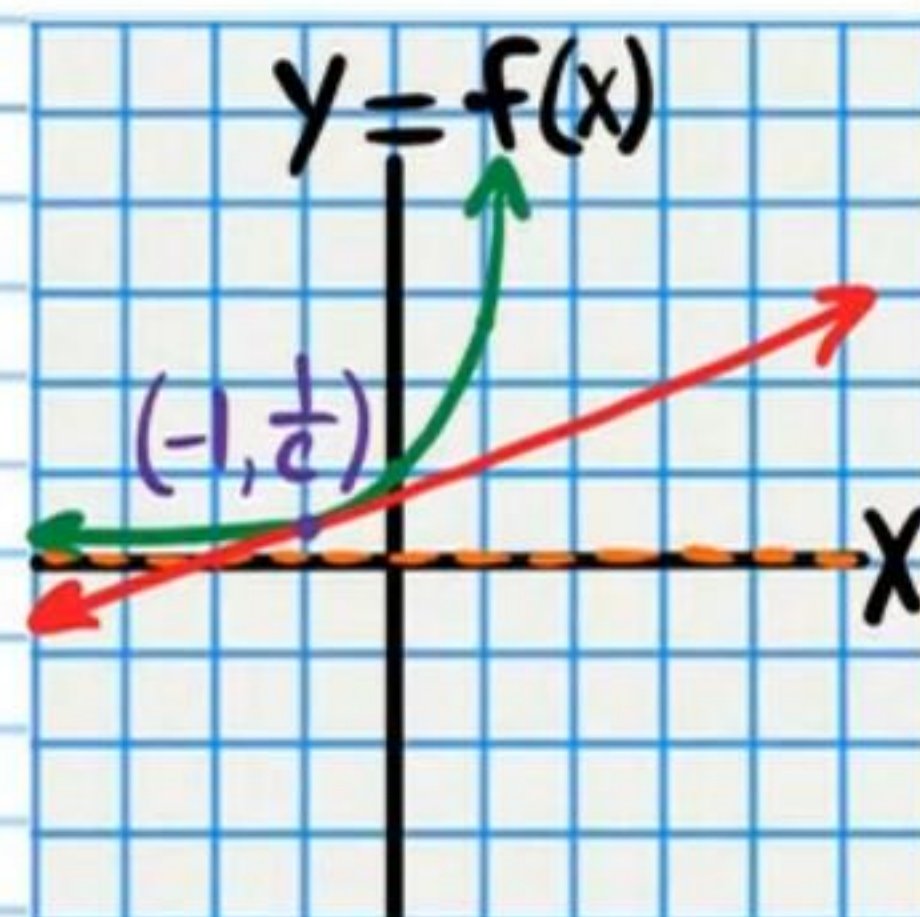
① $f(x) = 2(x-1)^2$, $(1\frac{1}{2}, \frac{1}{2})$



② $f(x) = \cos x$, $(\frac{3\pi}{4}, -\frac{\sqrt{2}}{2})$



③ $f(x) = e^x$, $(-1, \frac{1}{e})$



Find an equation of the line tangent to the graph of each function at the indicated x -value.

④ $f(x) = 5^{-3x-1}$, $x = \frac{1}{3}$

Finding the y -value of the point of tangency.

Finding the slope of the tangent line.

Finding the Equation

⑤ $f(x) = \sqrt[3]{x} + 4$, $x = -27$

Finding the y-value of the point of tangency.

Finding the slope of the tangent line.

Finding the Equation

⑥ $f(x) = \log_3 x$ $x = 3$

Finding the y-value of the point of tangency.

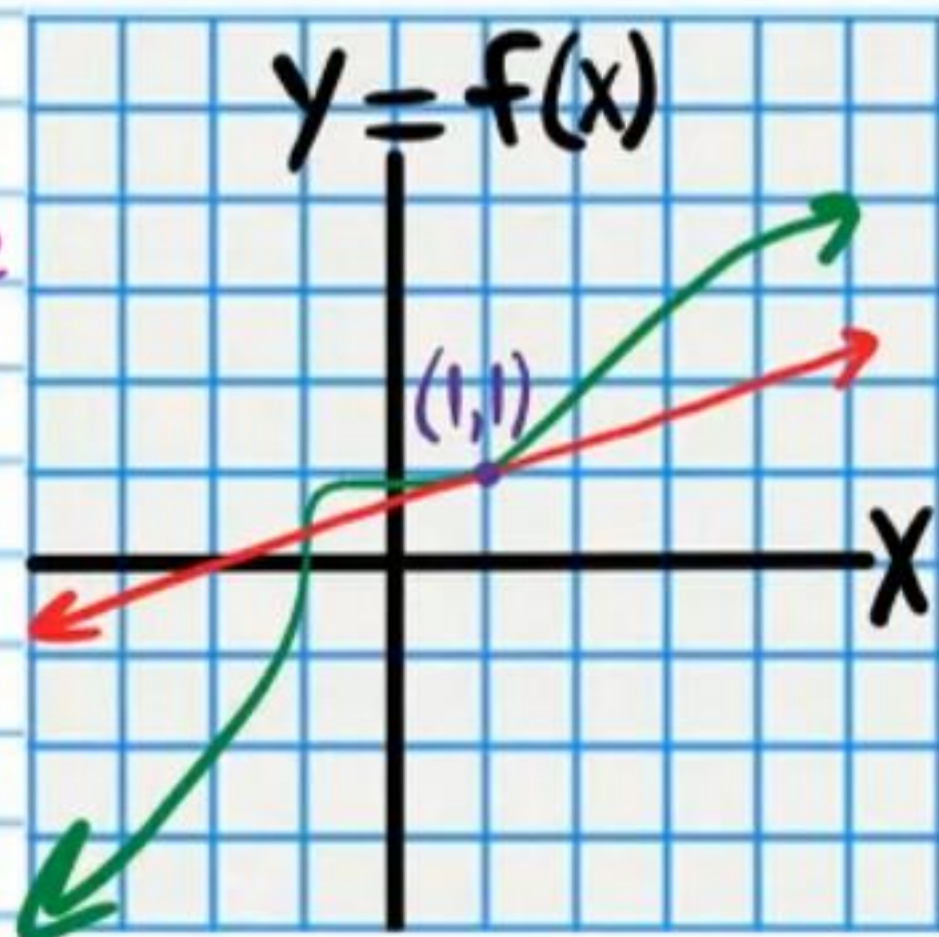
Finding the slope of the tangent line.

Finding the Equation

Use implicit differentiation to find an equation of the line tangent to the graph of each relation at the indicated point.

⑦ $1 + x^3 = 2y^5$, $(1, 1)$

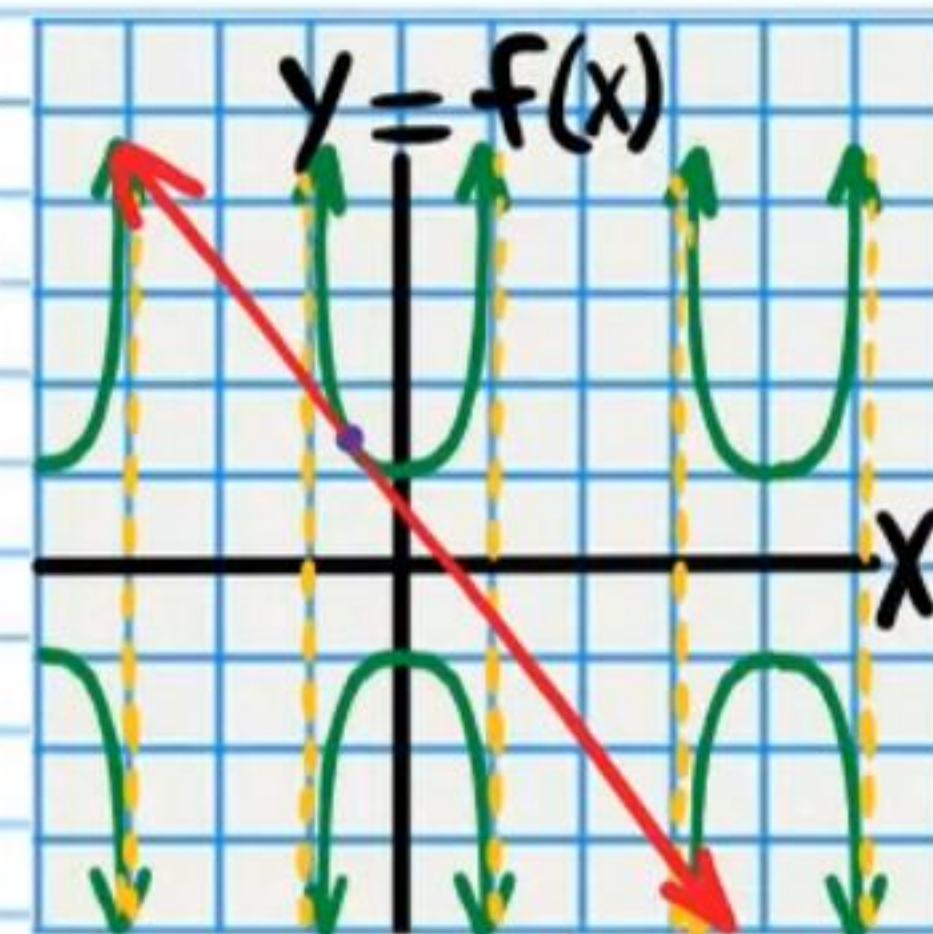
Finding the Slope of the Tangent Line



Writing the Equation of the Tangent Line

⑧ $y^2 = \sec x$ at $(-\frac{\pi}{3}, \sqrt{2})$.

Finding the Slope of the Tangent Line



Writing the Equation of the Tangent Line

Use implicit differentiation to find an equation of the line tangent to the graph of each relation at the indicated x -value or y -value.

⑨ $5x - \sqrt{y} = 2, \quad x = 1$

Finding the y -value of the point of tangency.

Finding the slope of the tangent line.

Finding the Equation

⑩ $x^3 = 1000\sqrt{3} \tan y, \quad y = \frac{7\pi}{6}$

Finding the x -value of the point of tangency.

Finding the slope of the tangent line.

Finding the Equation

⑪ $4x^3 = \ln y - 2, \quad X = 2$

Finding the y-value of the point of tangency.

Finding the slope of the tangent line.

Finding the Equation

Homework Answers

① $y - \frac{1}{2} = 2(x - \frac{3}{2})$

② $y + \frac{\sqrt{2}}{2} = -\frac{\sqrt{2}}{2}(x - \frac{3\pi}{4})$ or $y + \frac{1}{\sqrt{2}} = -\frac{1}{\sqrt{2}}(x - \frac{3\pi}{4})$

③ $y - \frac{1}{e} = \frac{1}{e}(x + 1)$

④ $y - \frac{1}{25} = -\frac{3 \ln 5}{25}(x - \frac{1}{3})$

⑤ $y - 1 = \frac{1}{27}(x + 27)$

⑥ $y - 1 = \frac{1}{3 \ln 3}(x - 3)$

⑦ $y - 1 = \frac{3}{10}(x - 1)$

⑧ $y - \sqrt{2} = -\frac{\sqrt{6}}{2}(x + \frac{\pi}{3})$

⑨ $y - 9 = 30(x - 1)$

⑩ $y - \frac{7\pi}{6} = \frac{3\sqrt{3}}{40}(x - 10)$

⑪ $y - e^{34} = 48e^{34}(x - 2)$